

Augstin Stark

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Early-19th-century **printed** sunspot records associated with **Augustin Stark** (and occasional contemporaries). Primarily narrative descriptions with day-level sunspot activity.

Coverage

~**1813–1835** (e.g., entries from 1817). Cadence is uneven with weather gaps; authorship may include adjacent contributors (e.g., “Hauptmann” drawing near Stark text).

Why it matters (FARSuN)

- Strengthens daily constraints in the **Dalton/post-Dalton** era before dense Schwabe records.
- Adds **NG/NS** (and spotless days) for calibration chains used in **ISN V3.1/V4**.
- Limb-distance and morphology cues support **hemispheric hints** and qualitative QC even without precise positions.

- Being **printed**, it is amenable to **OCR + prompt-guided extraction** (faster than HTR).

Present Status

- **Identified & sourced**; materials located in German digital libraries and shared drives.
- **Pilot extractions** via a Shiny/LLM-assisted interface (initially for Gruithuisen) show **good but inconsistent** results; further tuning needed.
- Listed as “**to be examined**” in project tracking; student trials done; workspaces prepared.

What the data look like (content & structure)

- **Per-date narrative**: morphology (“depressions/openings,” “chains”), distances from limb (/), weather notes.
- **Counts**: implicit/explicit **NG**, **NS**; **Wolf number** derivable ($W=10 \cdot NG+NS$).
- **Metadata spine**: date/time, observer, source (journal/volume/page), method/instrument if given, hemisphere hint, QC flags.

Digitization & QC

- **OCR** for Fraktur/antiqua + **rule-based normalization** of units and terms.
- **Prompt-guided extraction** with few-shot examples; codify “narrative → counts” rules (e.g., chains vs discrete spots).
- **Dual-annotator validation** and **adjudication**; derive inter-rater stats to seed per-day uncertainties.
- **Cross-observer checks** against Pastorff, Gruithuisen, Herschel to build **k-factor priors**.
- **Provenance**: retain page images/snippets, hashes, and versioning.

Accessibility

- Working copies/notes in internal wiki & drives.
- Post-QC ingestion to **ROB VO/TAP** with FARSuN's **Drawings / Groups / Conditions** schema; **FAIR** compliant metadata.

Near-Term To-Dos (practical)

1. Finalize **bibliography & page map**; disambiguate non-Stark inserts.
2. Publish a **prompt playbook** + error taxonomy; standardize extraction flags.
3. Align with contemporaries; estimate $\Delta\text{NG}/\Delta\text{NS}$ and scaling priors.

Path to ISN V3

- Integrate daily **NG/NS + uncertainties** as an **informative node** in calibration chains.
- Use Stark as a **procedural rehearsal** for V3 (expanded uncertainty & provenance).
- Release notes to include cadence limits, textual ambiguities, and multi-observer considerations.

Stark (1813–1835) is a high-value, printed dataset that can be reliably normalized with OCR+rules+dual-QC to support ISN V3.